

Predicting Battery State of Health and Remaining Useful Life using Explainable Machine Learning

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Abstract—Lithium-ion battery State of Health (SoH) and Remaining Useful Life (RUL) cannot be measured directly in battery management systems. This paper presents a dual-domain machine learning pipeline applied to the NASA PCOE Battery Dataset (B0005–B0018), following the CRISP-DM methodology [1]. Part 1 extracts 17 physics-informed & statistical features from discharge time-series and compares Ridge, Random Forest, and XGBoost regression under a cross-battery holdout, achieving a best SoH RMSE of 0.0041 Ah. SHAP explainability analysis identifies cumulative energy throughput and internal resistance as the dominant degradation drivers. Part 2 replicates and extends Zhao et al. [2] by extracting six health factors from charging curves, forecasting their trajectories with a multi-output LSTM, and mapping predicted factors to SoH via XGBoost. Post-hoc bias correction at 75th percentile (OOF residuals) reduces SoH RMSE from 0.0594 to 0.0434 Ah. Cross-domain SHAP analysis reveals that internal resistance influences capacity fade across independent measurement protocols, directly motivating a causal model as future work.

Index Terms—State of Health, Remaining Useful Life, LSTM, XGBoost, SHAP, CRISP-DM, battery prognostics, Quantile regression, explainable ML

I. BUSINESS UNDERSTANDING

Lithium-ion batteries degrade silently through repeated charge-discharge cycling. A battery management system (BMS) has no direct SoH sensor and observes only voltage, current, and temperature. Premature replacement wastes cost while delayed replacement risks unexpected failure. This poses the question “*can SoH and RUL be predicted accurately with explainability from signals already present in a BMS?*”

This project formulates two complementary regression tasks:

- 1) **Part 1 (SoH prediction + RUL)**: from discharge-curve features, using current-cycle signals to estimate current-cycle capacity.
- 2) **Part 2 (HF forecasting + SoH + RUL)**: from charging-curve health factors, enabling future-cycle trajectory prediction.

Cross-battery holdout. the models train on B0005, B0006, B0007 and are separately evaluated on B0018. This ensures generalization to an unseen cell setting a realistic BMS deployment scenario.

Explainability as a deliverable. SHAP values are computed for all models to identify physically grounded relationships steering the battery capacity. Cross protocol features

(Discharge and Charge domain) are compared using SHAP to assess if they are driven by the same underlying characteristics.

II. DATA UNDERSTANDING

A. Dataset

The NASA PCOE Battery Repository [3] provides multiple LiCoO₂ cells and four common cells (B0005 - B0018, nominal capacity 2.0 Ah) with similar test conditions are utilized for SoH prediction. They are discharged at constant 2 A to a 2.7 V cutoff at 24°C ambient, run to end-of-life (EOL) defined as 30% fade (1.4 Ah). Each cell yields approximately 168 discharge cycles and a matching sequence of charge and impedance cycles, producing approximately 200,000 raw timesteps across 636 discharge cycles.

B. Key EDA Findings

Non-monotonic degradation. Capacity curves exhibit local recovery bumps (capacity regeneration) caused by lithium redistribution during rest periods. This invalidates naive interpolation and motivates non-linear features. B0006 degrades very deeply (SoH: 0.58) while B0007 degrades slowly.

Voltage curve evolution. Discharge plateau duration shortens visibly with cycle number. The initial voltage drop at $t = 0$ increases with age which is a direct indication of rising internal resistance, justifying the use of `R_int_proxy` feature (Fig. 1).

Inter-battery variance. B0018 diverges from training batteries by up to 0.20 Ah around cycles 45-60, then partially converges mid-life. This non-stationary inter-battery divergence produces systematic early-cycle bias in cross-battery models, reported explicitly in residual analysis.

Split strategy validation. Three split strategies were compared using the same XGBoost model to quantify evaluation inflation (Table I).

TABLE I
SPLIT STRATEGY COMPARISON

Strategy	Validity
A: Random shuffle	Leakage from future cycles
B: Within-battery 50/50	validation observes train battery cycles
C: Cross-battery holdout (Group K fold)	Generalization across battery

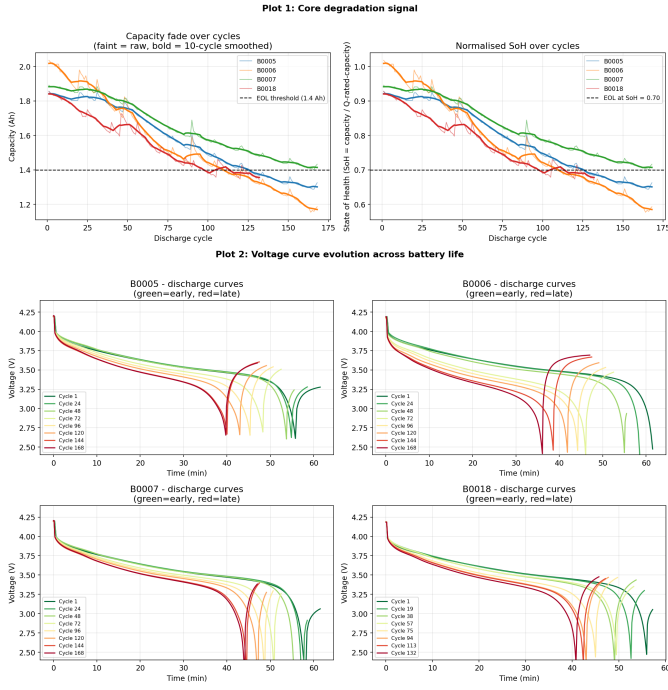


Fig. 1. EDA: capacity degradation and discharge curve evolution across battery life

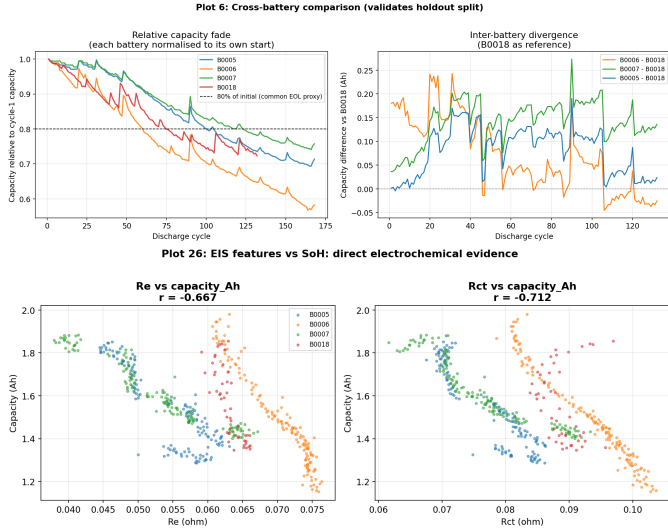


Fig. 2. EDA: Cross-battery Variance and capacity by EIS fields

III. DATA PREPARATION

A. Part 1: Discharge Feature Engineering

Each discharge cycle (300 time steps) is collapsed to 17 scalar features (Table VII) divided into two categories:

Statistical features: discharge duration, voltage mean/std/slope, voltage at 80% of discharge, temperature rise.

Physics-informed features: internal resistance proxy ($R_{int} \approx dV/dI$ at cycle start), cumulative charge (Q_{cum}),

energy throughput (E_{Wh}), temperature-compensated capacity (Q_{norm}), incremental capacity peak height and voltage (dQ/dV). Smoothened R_{int} and flag to capture initial bathtub curve where internal resistance drops. Lastly, all batteries demonstrate increased temperatures across cycles warranting the use of rise in temperature as a feature.

Rolling features (`rolling_mean_5`, `rolling_std_5`) are computed with a `shift(1)` offset to prevent target leakage the rolling mean for cycle n uses only cycles $n-5$ to $n-1$.

The heatmap directly measures the correlation with capacity and across features (Fig. 3). Autocorrelation concerns are internally handled by tree based methods due to single node split. All features pass through a `sklearn Pipeline` (StandardScaler $\rightarrow$ model), fit exclusively on training batteries to prevent leakage.

B. Part 2: Charge Health Factor Extraction

Following Zhao et al. [2], six health factors (HF1-HF6) are extracted from the CC - CV charging curve per cycle. Spearman correlation (simplified from paper's MIC analysis) selects HF1, HF2, HF3, HF6 (Table II) as input to the LSTM.

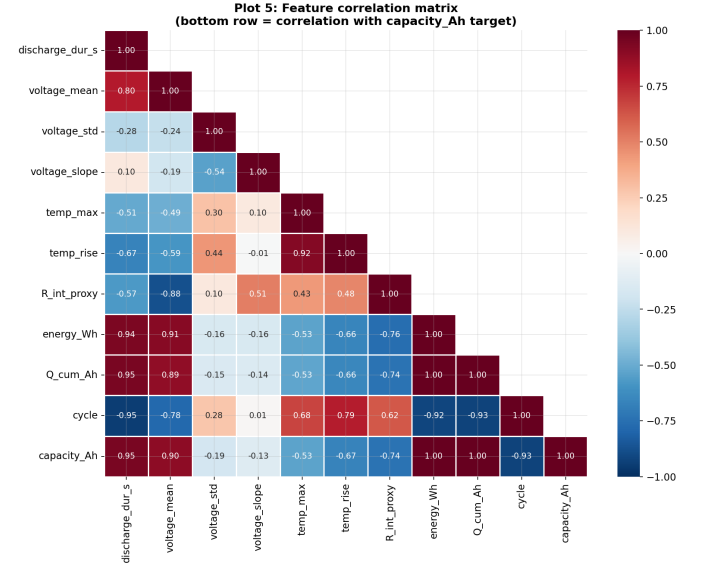


Fig. 3. Correlation Heatmap: Features and Capacity

TABLE II
HEALTH FACTOR SELECTION VIA SPEARMAN CORRELATION

HF	Description	Spearman r	Selected
HF1	CC charge time 3.9 $\rightarrow$ 4.2 V	0.935	✓
HF2	Voltage rise within 600 s	-0.919	✓
HF3	CV current drop within 900 s	0.598	✓
HF4	Exp. fit floor (I_{∞})	-0.172	×
HF5	Exp. fit amplitude	0.030	×
HF6	RC time constant	-0.871	✓

IV. MODELING

A. Part 1: Discharge Domain Models

Three models are trained and evaluated with GroupKFold cross-validation (3 folds, grouped by battery ID, train on B0005, B0006, B0007 test on B0018):

- **Ridge Regression** (baseline): L_2 -regularized linear model ($\alpha = 1.0$). Coefficients are directly interpretable, and features proportionally influence the target variable. SHAP via `LinearExplainer` provides the feature rank and direction.
- **Random Forest**: 200 trees, `min_samples_leaf=2`, `max_depth=10`.
- **XGBoost**: 300 estimators. Tree based SHAP - feature rank and feature engineering statistical features (Fig. 4).
- The hyperparameters are tuned using `GridSearchCV` with GroupKFold validation.

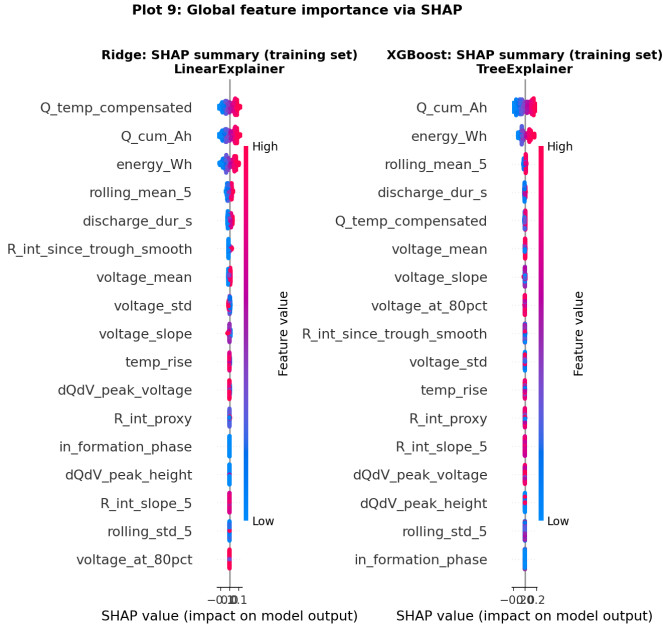


Fig. 4. Ridge and XGBoost SHAP analysis.

B. Part 2: LSTM + XGBoost Pipeline

The two-stage pipeline follows Zhao et al. [2] with XGBoost replacing GPR:

- **Stage 1: Multi-output LSTM**: A 2-layer LSTM (hidden size 32, dropout 0.3, Adam $\text{lr}=0.0005$) predicts all four HF values at cycle $n + 1$ from a sliding window of the last 10 cycles. Cross battery validation split (train on 2 batteries and validate on the remaining one). Dropout and gradient clipping prevent overfitting. Early stopping is applied and mean `MAX_EPOCH` across validation is used for full model training (Fig. 5).
- **One-step vs. rollout**: One-step predictions (true history window) demonstrate captured trend while auto-

regressive rollout pipeline provides a deployment simulation. Rollout suffers from compounding variance (Fig. 6).

- **Stage 2: XGBoost on predicted HF**: XGBoost trained on HF1, HF2, HF3, HF6 from training batteries maps predicted HF values to SoH. The cycle index is explicitly excluded to prevent target leakage.
- **Post-hoc calibration**: Raw predictions exhibit systematic under-prediction on B0018 due to inter-battery capacity offset and LSTM error propagation. The OOF mean shift, Q75 / Q90 residual shift, and XGBoost quantile regression methods mitigate bias (Fig. 7). The OOF mean shift downgrades performance as training batteries show bias in opposite-direction. 75th quantile residual shift achieves lowest bias. Native quantile XGBoost underperforms relative to post-hoc shift although provides a robust path for deployment pipeline.

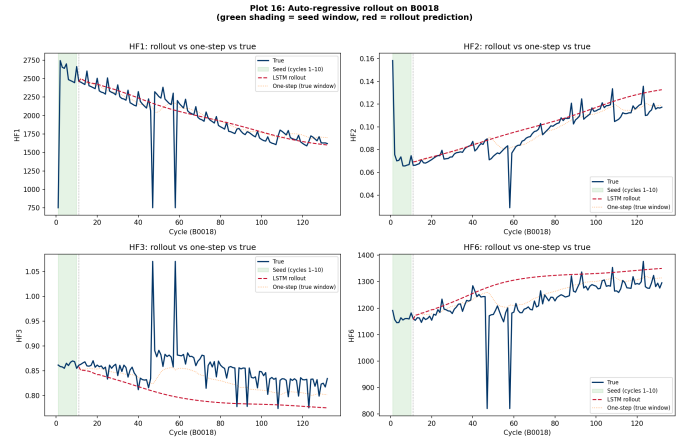


Fig. 5. LSTM predictions across horizon.

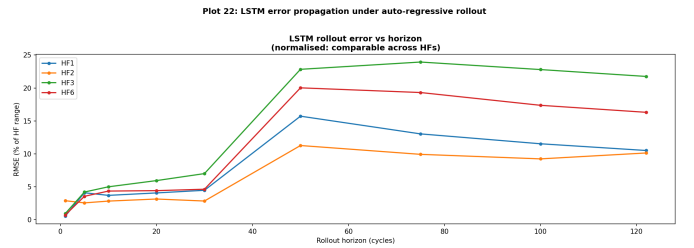


Fig. 6. LSTM error propagation under auto-regressive rollout. All HF6 converge to 15-22% of their operating range by horizon 50, driven by anomalous B0018 cycles 45-60.

V. EVALUATION

A. SoH Prediction

Table III summarizes SoH RMSE across all models along with respective protocol on B0018. Discharge-domain models slightly outperform charge-domain models as they use true measured values and discharge measurements directly

associate with same event being predicted. In charge domain calibrated model (Q75) offers lowest error.

The three models have low rmse despite significant inter-battery variance, implying that variance is a static offset and cumulative features (Q_Cum_Ah , $energy_Wh$) are efficiently encoding the capacity trajectory. This is in line with normalized capacity-fade (Fig 2) that exhibits closer relations between batteries.

The cross-protocol results are not directly comparable and address different prediction problems. The comparison is conceptual and limited to underlying mechanisms. Charging protocols are standardized and stable across operating conditions, part II pipeline mainly serves as a BMS deployment sample.

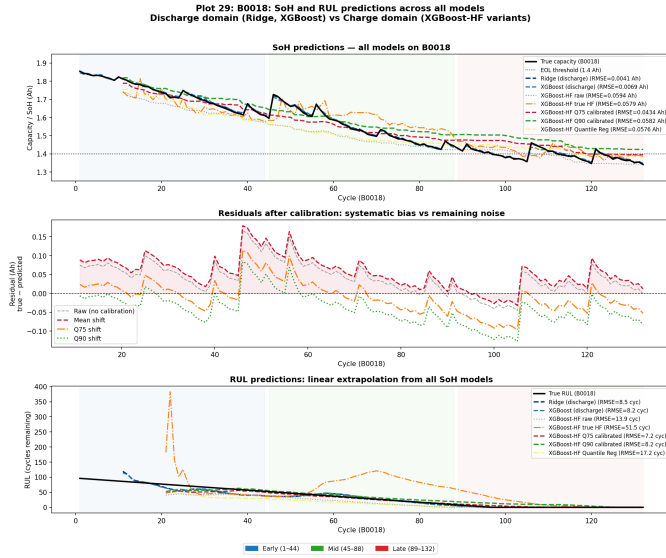


Fig. 7. SoH (with post-hoc calibration) and RUL predictions across all models on B0018.

TABLE III
SoH RMSE ON B0018 HOLDOUT

Model	RMSE (Ah)	Protocol
Ridge regression	0.0041	Discharge
XGBoost (discharge)	0.0069	Discharge
XGBoost-HF Q75 calibrated	0.0434	Charge (cross-bat.)
XGBoost-HF true HF	0.0579	Charge (cross-bat.)
XGBoost-HF raw	0.0594	Charge (cross-bat.)
XGBoost-HF Quantile Reg (0.75)	0.0576	Charge (cross-bat.)

Residual analysis on discharge model.

- Ridge shows a systematic sign pattern. Early cycles are diverse while late cycles trend negative, meaning Ridge over-predicts as the battery ages. (Fig. 8)
- RF and XGB under-forecasts capacity as degradation accelerates. This is the fingerprint of overfitting on the recovery bump patterns of training batteries, validating the inter-battery offset seen in EDA.
- Across the three predicted-vs-actual plots, points cluster on the diagonal, confirming the models remain reliable

near EOL where accuracy matters for replacement decisions, despite their mid-life residual structure.

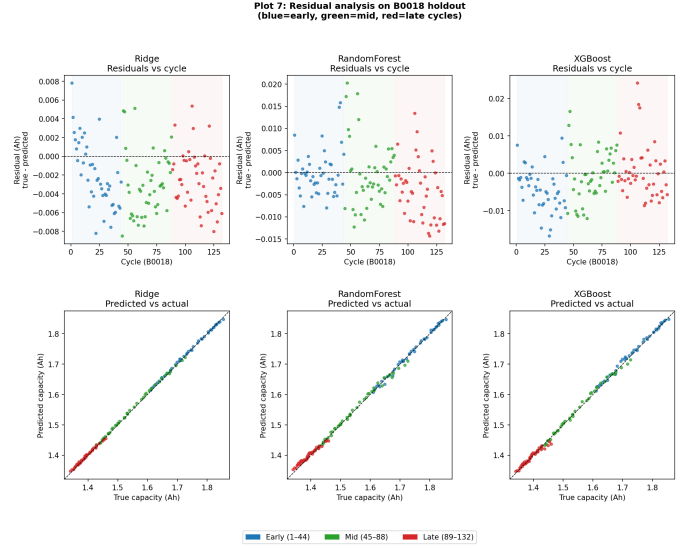


Fig. 8. Residual Analysis on Discharge models.

B. RUL Prediction

Linear extrapolation from each model's SoH trajectory yields the RUL estimates in Table IV. Discharge model (XGBoost) achieves lower rmse relative to raw LSTM+XGBoost model. Post-calibration with 75th percentile of OOF residuals achieves the best approximation of RUL curve across the life-cycle of battery (Fig. 7).

True HFs produce high RUL error (51.49 cycles) due to anomalous B0018 experimental cycles (45–60). LSTM smoothing reduces the error to 13.88 cycles, a fundamental trade-off between point accuracy and trajectory stability.

TABLE IV
RUL RMSE ON B0018 (LINEAR EXTRAPOLATION)

Model	Early (1-44)	Mid (45-88)	Late (89-132)	Overall
Ridge (discharge)	14.18	6.47	0.94	8.45
XGBoost (discharge)	13.59	6.44	1.29	8.19
XGBoost-HF raw	23.48	13.59	2.15	13.88
XGBoost-HF true HF	71.46	59.09	21.68	51.49
XGBoost-HF Q75	14.72	2.59	2.53	7.18
XGBoost-HF Q90	10.77	5.69	8.70	8.20

C. Explainability: Cross-Domain SHAP Convergence

SHAP analysis on both domain models reveals convergent evidence for a two-pathway degradation model (Fig. 9):

Discharge domain: $energy_Wh$ and Q_cum_Ah dominate. They act as capacity proxies and offer less noisy collinear substitutes for R_int_proxy .

Charge domain: $HF6_rolling_mean5$ (RC time constant trend) and $HF1$ (CC charge time) dominate. Both measure how internal resistance affects the charging process as

higher resistance lengthens CC charging and increases the RC time constant.

Convergence interpretation: As batteries age, internal resistance increases. From the discharge side, this reduces energy deliverable per cycle ($energy_Wh \downarrow$). From the charge side, this lengthens the time constant of CV current decay ($HF6 \uparrow$) and slows CC charging ($HF1 \uparrow$). Both pathways measure the same electrochemical mechanism from independent experimental protocols.

Degradation rate correlation reinforces this: $HF1$ evolution rate correlates with capacity degradation rate at $r = 0.997$ ($p = 0.003$) across all four batteries indicating some (hypothesis pending more samples) cross-domain link. $HF6$ shows consistent direction ($r = -0.882$) but does not reach significance at $n = 4$.

EIS corroboration: (Fig. 2) Impedance spectroscopy measurements show R_{ct} (Spearman $r = -0.70$) and R_e ($r = -0.64$) negatively correlated with capacity, providing independent electrochemical confirmation that resistance increase accompanies capacity fade [4]. EIS features are omitted due to sparse availability (48 of 132 B0018 cycles).

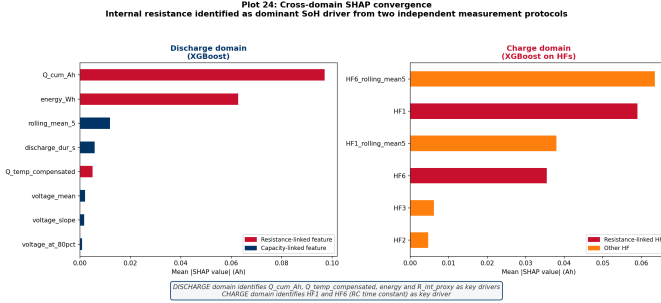


Fig. 9. Cross-domain SHAP convergence. Discharge domain (left) identifies cumulative throughput features; charge domain (right) identifies RC time constant and CC charge time. Both implicate internal resistance as the primary SoH driver.

VI. DEPLOYMENT & FUTURE WORK

A. Deployment Considerations

The discharge-domain ridge (smaller cohort) or XGB model (lower RUL rmse & flexible tuning) can be adopted for BMS applications where discharge telemetry is available as it eliminates sequence modeling.

The charge-domain pipeline is preferable when *future* SoH is needed from standardized charging sessions for fleet BMS where discharge conditions vary. The raw LSTM+XGBoost RUL predictions is operationally justified under conditions where conservative predictions are safer, as premature replacement is costly but unexpected failure could be dangerous.

A recalibration frequency of 30-40 cycles is recommended for the LSTM rollout, based on the error propagation analysis showing RMSE stabilization at horizon 50.

B. Limitations

Results are evaluated on a single test battery (B0018). The cross-battery generalization gap is partially explained by

B0018's lower starting capacity relative to training batteries. A larger battery cohort is necessary to improve evaluation robustness.

The health-factor forecast is seeded from only the first 10 true cycles and then self-feeds for the remaining 120+ cycles. For longer cycles, LSTM prediction errors are cumulative. The low rmse of Q75 calibration is only retrospective, hence quantile regression offers an intermediate solution for deployment scenario. Linear extrapolation for RUL is a simplified baseline assumption, and actual deployment requires non-linear modeling.

C. Causal Roadmap

The cross-domain SHAP convergence and degradation rate correlations motivate a DAG-based causal model as possible follow up:

$$\text{Aging} \rightarrow R_{\text{int}} \uparrow \rightarrow \begin{cases} E_{\text{Wh}} \downarrow & (\text{discharge}) \\ HF6 \uparrow, HF1 \uparrow & (\text{charge}) \end{cases} \rightarrow \text{SoH} \downarrow \quad (1)$$

This can utilize the present correlation model as a precursor, establishing which variables carry signal before asking which are causal.

VII. CONCLUSION

This work demonstrates that battery SoH and RUL can be predicted accurately with explainability from both discharge and charging signals using tabular ML methods. Discharge-domain Ridge regression achieves RMSE 0.0041 Ah outperforming more complex architectures under a stricter evaluation protocol.

The LSTM+XGBoost charge-domain pipeline replicates and extends Zhao et al. [2], replacing GPR with a calibrated XGBoost and adding SHAP explainability absent from the original paper.

Cross-domain SHAP analysis reveals that internal resistance, measured as cumulative throughput decline from discharge curves and RC time constant growth from charging curves, is the convergent driver of SoH degradation across independent experimental protocols. This convergence constitutes the main finding and directly motivates a causal model for future work.

Explainability KPIs, especially in physical systems and digital twins, are just as relevant as quantifiable metrics.

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GLOSSARY OF PHYSICS AND ELECTROCHEMICAL TERMS

Term	Definition
<i>Battery States & Targets</i>	
SoH	State of Health. Ratio of current discharge capacity to nominal capacity: $\text{SoH} = Q_n/Q_0$. Equals 1.0 for a fresh cell; EOL is defined at $\text{SoH}=0.70$ (30% fade from 2.0 Ah to 1.4 Ah).
SoC	State of Charge. Fraction of available capacity currently stored in the cell. Distinct from SoH, a degraded cell can be fully charged ($\text{SoC}=1$) yet have low SoH.
RUL	Remaining Useful Life. Number of discharge cycles remaining before capacity falls below the EOL threshold (1.4 Ah).
EOL	End of Life. The cycle at which capacity first falls below the EOL threshold. Defined here as 30% fade from nominal capacity per NASA PCOE protocol.
<i>Electrochemical Mechanisms</i>	
SEI	Solid Electrolyte Interphase. A passivation layer that forms on the anode surface during early cycling. SEI growth consumes lithium irreversibly and increases charge transfer resistance (R_{ct}) over time.
R_{int}	Internal Resistance. Total ohmic and kinetic resistance of the cell. Comprises electrolyte resistance (R_e) and charge transfer resistance (R_{ct}). Rises monotonically with aging; directly reduces deliverable voltage and capacity.
R_e	Electrolyte Resistance. Ohmic resistance of the electrolyte and current collectors. Measured as the high-frequency intercept on the EIS Nyquist plot. Increases with electrolyte decomposition and lithium salt depletion.
R_{ct}	Charge Transfer Resistance. Resistance to lithium-ion transfer across the electrode (electrolyte interface). Measured as the Nyquist semicircle diameter. Increases with SEI growth and active material loss.
Intercalation	Process by which lithium ions insert into and extract from the layered electrode material (LiCoO_2 cathode, graphite anode) during charge and discharge. Repeated intercalation causes mechanical stress and capacity fade.
Capacity regeneration	Transient local increase in discharge capacity observed mid-life. Caused by partial reversal of side-reaction products during rest periods between test sessions. Produces non-monotonic capacity curves requiring rolling-window features.
LiCoO_2	Lithium Cobalt Oxide. Cathode material used in the NASA PCOE cells. Operates via lithium intercalation between cobalt oxide layers.
<i>Discharge-Domain Features</i>	
discharge_dur_s	Total discharge duration in seconds from cycle start to voltage cutoff at 2.7 V. Strong predictor of SoH (Pearson $r = 0.95$) because at constant 2 A current, duration directly encodes charge delivered.
R_int_proxy	Internal resistance proxy computed as $\Delta V/\Delta I$ over the first 10 timesteps of each discharge cycle. Approximates R_{int} from the transient voltage drop under applied load.
energy_Wh	Energy delivered per cycle: $E = \int V \cdot I dt/3600$ (Wh). Captures both voltage and current contributions to capacity, top SHAP feature in discharge domain.
Q_cum_Ah	Cumulative charge delivered per cycle: $Q = \int I dt/3600$ (Ah). At constant 2 A discharge, highly correlated with <code>discharge_dur_s</code> ($r \approx 1.00$).
Q_temp_compensated	Temperature-compensated capacity: $Q_{\text{norm}} = Q/(1 + \alpha(T_{\text{amb}} - T_{\text{ref}}))$ where $\alpha = 0.005^\circ\text{C}^{-1}$. Removes thermal drift from the capacity signal.
dQdV_peak	Incremental capacity peak height extracted from the dQ/dV curve over the 2.7-3.7 V window. Peaks correspond to electrode phase transitions; amplitude decreases and position shifts with aging.
temp_rise	Temperature increase within a discharge cycle: $T_{\text{rise}} = T_{\text{max}} - T_{\text{start}}$. Used in preference to T_{max} to cancel inter-battery ambient temperature offsets.
voltage_slope	Linear regression slope of terminal voltage over discharge time (V/s). Becomes more negative with aging as internal resistance accelerates voltage decay.
rolling_mean_5	Rolling mean of <code>capacity_Ah</code> over the preceding 5 cycles, computed with <code>shift(1)</code> to prevent target leakage. Captures local degradation trend through non-monotonic recovery regions.
R_int_since_trough_smooth & in_formation_phase	Accounts for initial trough in initial resistance due to SEI layer stabilization. δR_{int} with cumulative minimum and flag initial trough.

Term	Definition
<i>Charge-Domain Health Factors</i>	
HF1	Time for terminal voltage to rise from 3.9 V to 4.2 V during CC charging. Increases with R_{int} as higher resistance slows voltage rise rate. Spearman $r = 0.935$ with capacity.
HF2	Voltage increase within 600 s after reaching 3.9 V during CC charging. Decreases with aging as the cell charges more slowly. Spearman $r = -0.919$.
HF3	Current reduction within 900 s after entering CV phase. Reflects how quickly the cell accepts charge at fixed voltage. Spearman $r = 0.598$.
HF6	Time constant of exponential current decay during CV phase, fit as $I(t) = a + b e^{-t/\text{HF6}}$. Directly measures the RC time constant of the battery equivalent circuit; increases with R_{int} .
<i>Model & Evaluation Terms</i>	
CRISP-DM	Cross-Industry Standard Process for Data Mining. Six-phase iterative methodology: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, Deployment.
SHAP	SHapley Additive exPlanations. Game-theoretic framework assigning each feature a contribution value to each prediction. Mean SHAP provides global feature importance.
GroupKFold	Cross-validation strategy where fold assignments respect group boundaries (here: battery ID). Prevents cycles from the same battery appearing in both train and validation folds.
OOF residuals	Out-of-Fold residuals. Prediction errors on validation folds during cross-validation, used here to estimate the calibration offset for post-hoc bias correction.
Quantile calibration	Post-hoc additive correction equal to the 75th & 90th percentile of OOF residuals. Shifts predictions upward to correct systematic under-prediction from inter-battery capacity offset.
<i>Measurement Protocols</i>	
CC-CV	Constant Current – Constant Voltage charging protocol. Phase 1: charge at fixed current (1.5 A) until voltage reaches 4.2 V. Phase 2: hold at 4.2 V while current decays to 20 mA cutoff.
EIS	Electrochemical Impedance Spectroscopy. Applies a small AC perturbation across a frequency sweep (0.1 Hz to 5 kHz) to decompose internal resistance into R_e and R_{ct} components via Nyquist plot analysis.
Nyquist plot	Complex impedance plot ($-\text{Im}(Z)$ vs $\text{Re}(Z)$) produced by EIS. High-frequency intercept gives R_e ; semicircle diameter gives R_{ct} .
Cutoff voltage	Minimum terminal voltage at which discharge is halted. Set to 2.7 V for B0005–B0018 in this dataset. Determines the depth of discharge and affects the comparability of capacity measurements across battery groups.